

Piyush Kumar

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ABOUT

Computer Science undergraduate specializing in **Computer Networking** with a strong foundation in **Artificial Intelligence, Machine Learning, Data Structures, and System Design**. Experienced in developing **AI-driven applications, network-based systems and full-stack solutions** through research projects and IoT systems. Skilled in **Java, Python, ML algorithms, and networking technologies**, focused on building scalable and efficient engineering solutions.

EDUCATION

SRM Institute of Science and Technology	2023 – 2027
<i>B.Tech in Computer Science and Engineering (Computer Networking)</i>	CGPA: 9.83 / 10
T.N.B. College, Bhagalpur	2021
<i>Class XII (BSEB) – PCM</i>	85.4%
D.A.V. Public School, Bhagalpur	2019
<i>Class X (CBSE) – PCMB</i>	89.4%

EXPERIENCE

King Faisal University	Jun 2025 – Present
<i>Research Intern – AI/ML (Solar Energy Prediction)</i>	<i>Online, Saudi Arabia</i>
– Developed a hybrid regression-based solar energy forecasting system using 5+ years of real climatic data , improving prediction accuracy by 20% .	
– Implemented ensemble and Bi-LSTM models, reducing forecasting error (RMSE) by 15% through optimized feature engineering.	
SRM Institute of Science and Technology	May 2025 – Jul 2025
<i>Summer Research Intern – AI/ML</i>	<i>Kattankulathur, India</i>
– Enhanced image denoising pipelines for low-dose CT scans, improving signal-to-noise ratio and visual clarity by 20% .	
– Built ML-based pulmonary nodule detection models achieving 85%+ validation accuracy .	

PROJECTS

DSA Visualization Platform [Algorithm Visualizer]	Mar 2026 – Present
<i>Next.js, Data Structures & Algorithms, Supabase Auth, REST APIs, Vercel</i>	
– Developed an interactive platform to visualize Data Structures & Algorithms with step-by-step animations for sorting, searching, graphs, trees, linked lists, and dynamic programming.	
AI Email Reply Generator [Email Writer Assistant]	Jan 2026 – Feb 2026
<i>Java, Spring Boot, React.js, Gemini API, Render</i>	
– Designed and deployed a full-stack AI email automation system using Spring Boot with secure API integration and production-level error handling.	
Career Guidance Platform [Guidance Platform]	Sep 2025 – Dec 2025
<i>TypeScript, MongoDB, PostCSS, Google Maps API</i>	
– Built a scalable full-stack career guidance system using React and MongoDB with location-based service integration.	
Smart Home Automation with Jarvis [Demo]	Jan 2025 – Jun 2025
<i>Raspberry Pi, ESP8266, Python, IoT, Relay Module</i>	
– Implemented a voice-controlled smart home system with real-time monitoring.	

SKILLS

- **Programming Languages:** Java, Python, JavaScript
- **Web Development:** React.js, Next.js, HTML5, CSS3, Material UI
- **Backend & APIs:** Spring Boot, RESTful APIs, MongoDB, MySQL, PostgreSQL
- **AI / Machine Learning:** Machine Learning, TensorFlow, Scikit-learn, Large Language Models (LLMs), Prompt Engineering, Retrieval Augmented Generation (RAG)
- **Cloud & DevOps:** AWS, Oracle Cloud Infrastructure (OCI), OCI Generative AI Services, Render, GitHub
- **Networking & Security:** Computer Networking, SD-WAN, Cisco DNA Center, Network Security
- **Tools & Platforms:** Git, Cisco Packet Tracer
- **Hardware / IoT:** Raspberry Pi, ESP8266, Arduino, Sensors

CERTIFICATIONS

- Oracle Certified Professional – Generative AI
- NPTEL – Programming in Java

AWARDS AND SCHOLARSHIPS

Research Nexus Certificate [Certificate]	3rd Position
<i>SRM Institute of Science and Technology</i>	

Certificate of Merit [\[Credential\]](#)
Department of Physics and Nanotechnology, SRMIST
Performance-Based Scholarship [\[Verification\]](#)
SRM Institute of Science and Technology

1st Position
2 Times (2024, 2025)

LANGUAGES

English (Working Proficiency) Hindi (Native) German (Elementary)